

D²TMIL: Disentangled Dual-Task Multiple Instance Learning for 1p/19q Status Prediction from Whole Slide Images

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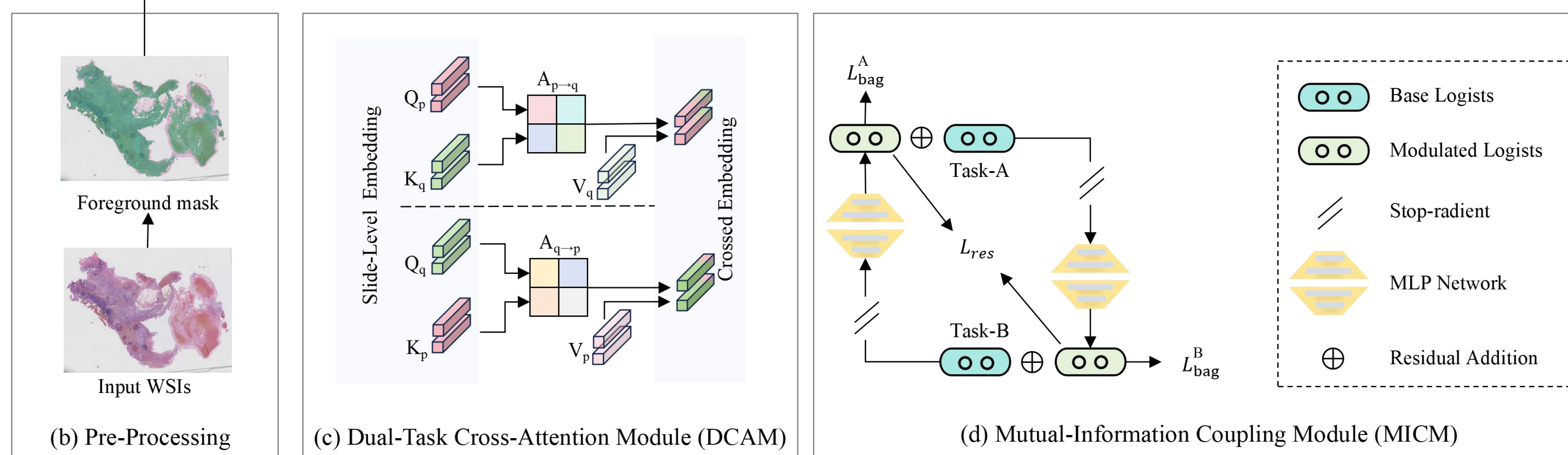
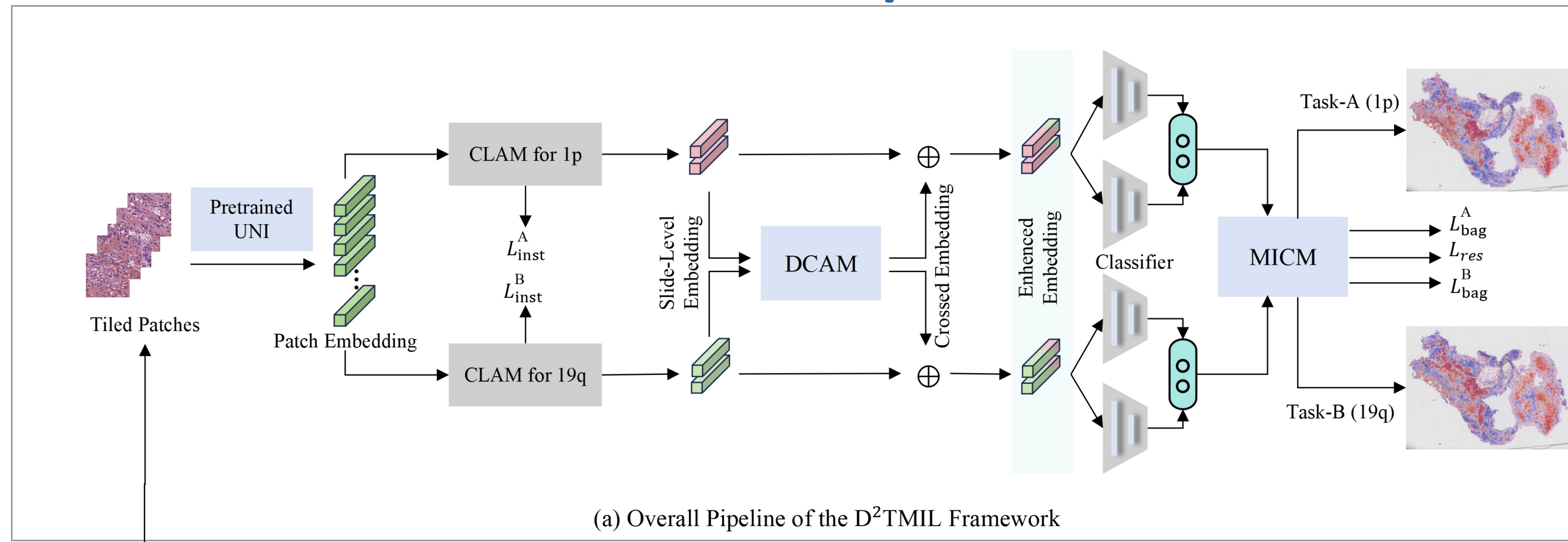
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Introduction

- Existing deep learning methods oversimplify 1p/19q prediction as binary classification, discarding individual chromosomal-arm deletion information that carries prognostic value (e.g., retained 1p with deleted 19q indicates poorer prognosis).
- They also ignore the heterogeneity of 1p/19q deletion patterns and the association between the two arms.

We propose a disentangled dual-task framework that models 1p and 19q deletions separately and infers codeletion via a joint probability distribution.

Overview of the Proposed Method



Each WSI is patched and encoded by UNI. Following CLAM, 1p/19q codeletion is decoupled into 1p and 19q deletion tasks, coupled via DCAM and MICM modules. Their joint probability yields the codeletion prediction and lesion heatmap.

Methodology

Disentangled Dual-Task MIL

- We decouple 1p/19q codeletion prediction into two correlated MIL subtasks, i.e., 1p deletion and 19q deletion. For each task $t \in \{A, B\}$, a multi-branch attention module generates class-wise attention weights $A_i^t \in \mathbb{R}^{C \times M_i}$ and aggregates instance features into class-specific bag representations:

$$m_{i,c}^t = \sum_{j=1}^{M_i} a_{i,c,j}^t h_{i,j}, \quad c = 1, \dots, C.$$

Dual-Task Cross-Attention Module (DCAM)

- We concatenate the two task-specific representations into a joint token sequence $T_i = [M_i^A; M_i^B] \in \mathbb{R}^{2G \times D}$ and apply multi-head attention to enhance inter-task interaction:

$$T_i = \text{MHA}(T_i, T_i, T_i), \quad \text{Attention}(Q, K, V) = \text{softmax}(QK^\top / \sqrt{d_h})V.$$

- The attended tokens are split back and injected into the original representations via residual connections:

$$M_i^A = M_i^A + c_i^A, \quad M_i^B = M_i^B + c_i^B.$$

Mutual-Information Dual-Task Coupling Module (MICM)

- We perform bidirectional information propagation between the output distributions of the two tasks using lightweight nonlinear mappings with stop-gradient:

$$\Delta z_i^A = g_{B \rightarrow A}(\text{sg}(z_i^B)), \quad \Delta z_i^B = g_{A \rightarrow B}(\text{sg}(z_i^A)).$$

Methodology

Joint Prediction and Optimization

- The two branches predict individual chromosomal-arm deletion status, and the codeletion probability is estimated from their joint probability:

$$p_i^{co} = p_i^A(1) \cdot p_i^B(1).$$

- The overall optimization objective is:

$$\mathcal{L}_{\text{total}} = \alpha \mathcal{L}_{\text{bag}}^A + \beta \mathcal{L}_{\text{bag}}^B + \gamma_A \mathcal{L}_{\text{inst}}^A + \gamma_B \mathcal{L}_{\text{inst}}^B + \eta \mathcal{L}_{\text{res}}, \quad \mathcal{L}_{\text{res}} = \|\Delta z_i^A\|_2^2 + \|\Delta z_i^B\|_2^2.$$

Result

Quantitative evaluation

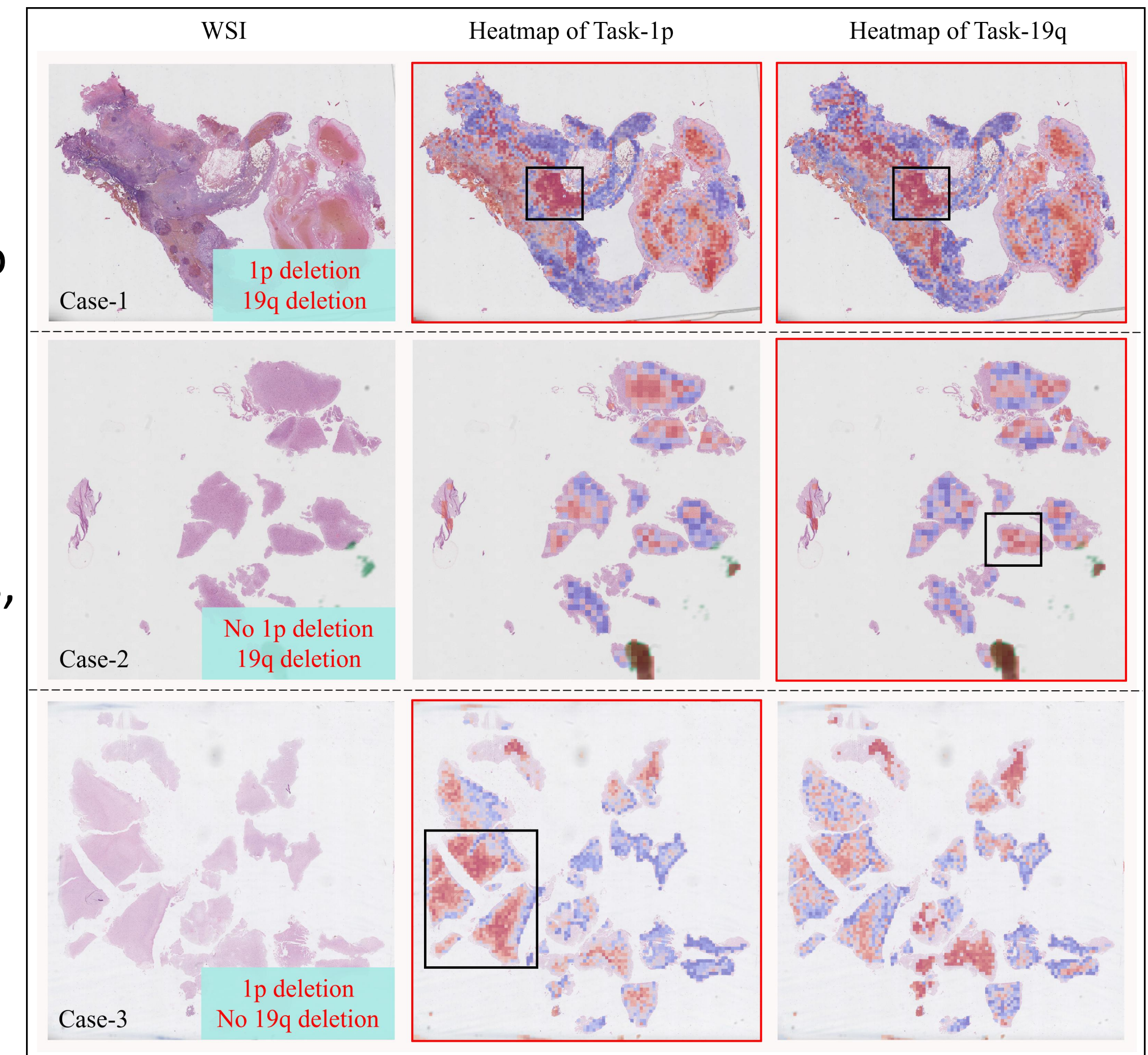
- CLAM serves as the "decomposition-only, no coupling" baseline.
- D²TMIL further improves performance over CLAM.

Method	1p			19q			Codeletion		
	ACC (%)	AUC (%)	F1 (%)	ACC (%)	AUC (%)	F1 (%)	ACC (%)	AUC (%)	F1 (%)
CLAM-SB	83.87 ± 8.22	92.72 ± 2.32	72.23 ± 10.36	87.10 ± 3.95	83.10 ± 1.49	67.46 ± 14.26	88.39 ± 3.68	97.20 ± 0.99	64.72 ± 14.31
CLAM-MB	87.74 ± 5.55	92.83 ± 3.68	78.16 ± 8.17	84.52 ± 2.41	83.45 ± 1.61	56.51 ± 19.01	89.68 ± 4.28	98.00 ± 1.19	73.68 ± 13.55
TransGNN	87.10 ± 3.53	94.02 ± 3.35	77.47 ± 5.70	87.10 ± 2.04	84.17 ± 1.34	67.84 ± 11.89	90.97 ± 1.29	97.73 ± 0.53	79.44 ± 2.39
Ours	87.74 ± 3.16	94.02 ± 2.98	78.44 ± 4.25	87.74 ± 3.76	85.00 ± 1.74	73.00 ± 6.40	93.55 ± 3.53	97.73 ± 1.61	84.10 ± 8.45

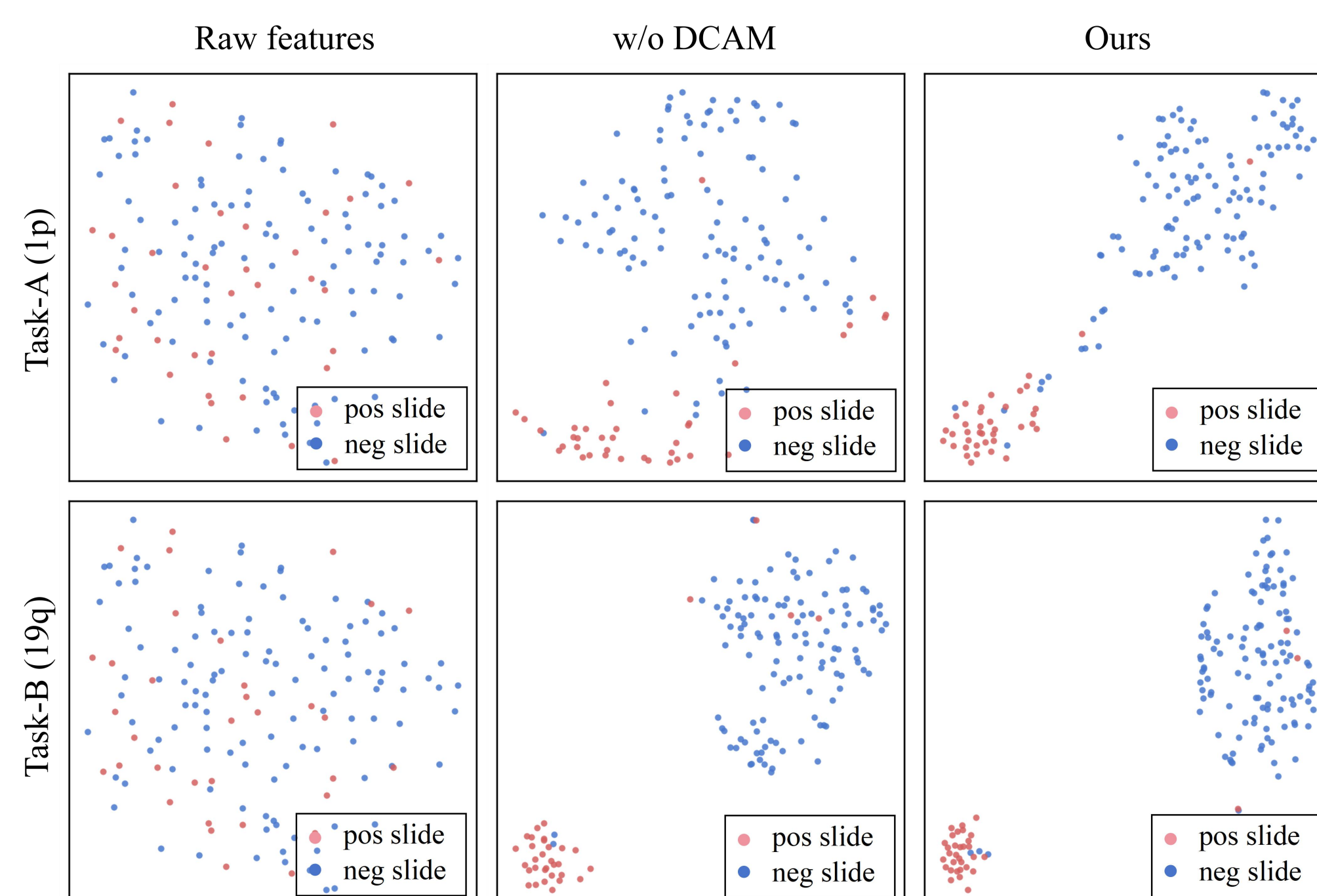
Qualitative evaluation

- Task-Specific Interpretability via Attention Heatmaps

The 1p and 19q branches attend to different ROIs in non-codeleted samples but similar ROIs in codeleted samples, demonstrating the task-specific interpretability of the decoupled design.



- Improved Feature Representation via t-SNE



a. Learned embeddings show clearer decision boundaries and stronger intra-class compactness.

b. Embeddings after introducing DCAM achieve improved inter-class separability.

Ablation Study

- Both Modules Are Necessary, and Bidirectional Coupling Outperforms Unidirectional Coupling

Method	1p			19q			Codeletion		
	ACC (%)	AUC (%)	F1 (%)	ACC (%)	AUC (%)	F1 (%)	ACC (%)	AUC (%)	F1 (%)
W/o DCAM	87.74 ± 4.74	91.20 ± 3.74	75.19 ± 7.51	85.16 ± 2.58	84.64 ± 2.95	55.17 ± 18.47	90.97 ± 1.29	96.00 ± 2.27	75.31 ± 5.49
W/o MICM	85.16 ± 3.87	93.91 ± 2.90	73.69 ± 6.20	86.45 ± 3.16	84.29 ± 1.28	72.17 ± 5.42	91.61 ± 4.38	98.00 ± 1.12	78.39 ± 11.46
W/o 1p→19q	85.81 ± 6.64	94.46 ± 3.00	75.43 ± 9.26	85.81 ± 5.98	83.45 ± 1.02	72.30 ± 7.20	89.03 ± 3.29	96.53 ± 1.60	75.84 ± 4.59
W/o 19q→1p	86.45 ± 5.55	94.67 ± 3.06	75.59 ± 6.74	86.45 ± 5.16	83.57 ± 0.89	72.16 ± 6.44	90.32 ± 3.53	97.33 ± 2.02	78.70 ± 5.34
Ours	87.74 ± 3.16	94.02 ± 2.98	78.44 ± 4.25	87.74 ± 3.76	85.00 ± 1.74	73.00 ± 6.40	93.55 ± 3.53	97.73 ± 1.61	84.10 ± 8.45